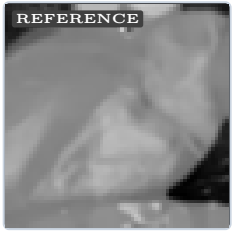
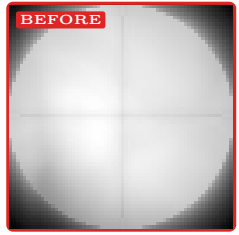
Clinical In Vivo CT and MRI Reconstructions: Conventional Slab-Clip vs. 3D AP-Bresenham

Reference Patient Slice BEFORE: Conventional Slab-Clip AFTER: 3D AP-Bresenham Artifact Error Map ($|\Delta I|$)

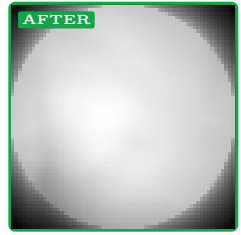
Clinical Abdominal CT



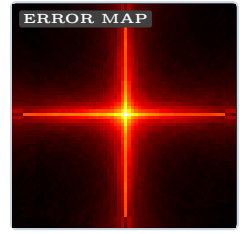
(a) In Vivo Patient CT
Axial Reference Slice
Clinical abdominal scan
255 × 178 HU Volume



(b) Slab-Clip + Reseeded
Continuous clip + round
Phase reset error (D_0)
92.2% corrupted | Streaks



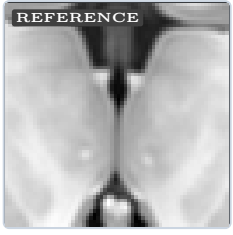
(c) 3D AP-Bresenham (Ours)
Exact Diophantine entry
Phase injection $D(k)$
0.0% defect | Bit-exact



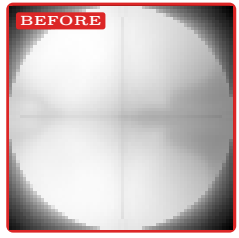
(d) Difference ($|\Delta I|$)
Conventional minus AP
Isolated boundary streaks
Max error = 15.94 cm^{-1}

$\Delta\mu$ error up to 15.94 cm^{-1} Exact Radon Integral Conservation (RMSE = 0.0000, PSNR = ∞)

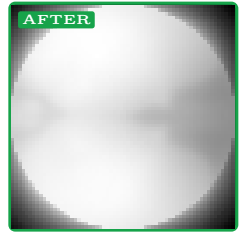
MNI-152 Brain MRI



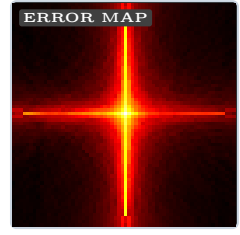
(e) MNI-152 Brain Template
Axial Reference Slice
Isotropic 1 mm^3 anatomy
207 × 256 T1 Volume



(f) Slab-Clip + Reseeded
Continuous clip + round
Phase-shift 2-voxel jump
90.2% corrupted | Step error



(g) 3D AP-Bresenham (Ours)
Exact Diophantine entry
Phase continuation $D(k)$
0.0% defect | Seamless



(h) Difference ($|\Delta I|$)
Conventional minus AP
Isolated boundary steps
Max error = 41.61